

Mohammad Refat

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RESEARCH INTERESTS

Computational astrophysics, Exoplanets, Time-series analysis, Inverse problems, Statistical inference, Scientific computing

EDUCATION

CUNY Graduate Center 2023–2025

M.S. in Astrophysics

Thesis: *Starspot Inference Using Light Curve Inversion Techniques*

CUNY Baccalaureate for Unique and Interdisciplinary Studies 2019–2022

B.S. in Computational Astrophysics, *cum laude*

Bernard M. Baruch College 2017–2019

RESEARCH EXPERIENCE

Starspot Inference with Light Curve Inversion Techniques Aug 2023 – Sep 2025

American Museum of Natural History

Advisor: Dr. Lucy Lu

- Developed computational approaches for starspot inference using light curve inversion techniques.
- Applied time-series analysis and probabilistic modeling to characterize stellar surface features.
- Implemented Python-based scientific computing workflows for astrophysical inference and model evaluation.

Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres May 2021 – Aug 2023

American Museum of Natural History

Advisor: Dr. Johanna Vos

- Investigated atmospheric variability in brown dwarfs and giant exoplanets through analysis of photometric light curves.
- Developed computational workflows for mapping cloud structures from observational time-series data.
- Applied quantitative modeling techniques to infer atmospheric properties from astronomical observations.

Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2 May 2020 – May 2021

Sloan Digital Sky Survey Faculty and Student Team

Advisor: Dr. Allyson Sheffield

- Analyzed APOGEE-2 spectroscopic data to identify candidate stellar members of the Jhelum stellar stream.
- Applied statistical and computational analysis methods to characterize stellar populations.
- Contributed to a refereed publication in *The Astrophysical Journal*.

Looking at Star Formation Through Chemistry Aug 2018 – May 2020

American Museum of Natural History

Advisor: Dr. Allyson Sheffield

- Analyzed stellar spectra to derive chemical abundances and characterize stellar populations.
- Processed and normalized echelle spectra and measured equivalent widths of spectral lines.
- Applied quantitative techniques to spectroscopic datasets.

Two-Point Statistics in the Star Forming ISM

Jun 2018 – Aug 2018

Center for Computational Astrophysics, Flatiron Institute

Advisor: Dr. Chang-Goo Kim

- Performed statistical analysis of metallicity correlations in star-forming regions.
- Investigated relationships among stellar metallicity, velocity, and star formation rate.

REFEREED PUBLICATIONS

Sheffield, Allyson A. et al. (May 2021). “Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2”. In: *The Astrophysical Journal* 913.1, p. 39. ISSN: 1538-4357. DOI: [10.3847/1538-4357/abee93](https://doi.org/10.3847/1538-4357/abee93). URL: <http://dx.doi.org/10.3847/1538-4357/abee93>.

ACADEMIC PRESENTATIONS

Talks

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| “Starspot Inference with Light Curve Inversion Techniques”
<i>CUNY Masters Graduation 2025</i>
Simons Foundation | 2025 |
| “Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres”
<i>240th American Astronomical Society Meeting</i>
American Astronomical Society | 2022 |
| “Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres”
<i>Center for Computational Astrophysics / City University of New York / American Museum of Natural History Symposium</i>
Center for Computational Astrophysics | 2021 |
| “Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2”
<i>American Museum of Natural History REU Symposium</i>
American Museum of Natural History | 2020 |
| “Where Were Stars Born in the Milky Way?”
<i>American Museum of Natural History REU Symposium</i>
American Museum of Natural History | 2019 |
| “Two-Point Statistics in the Star Forming ISM”
<i>Center for Computational Astrophysics REU Symposium</i>
Simons Foundation | 2018 |

Posters

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| “Starspot Inference with Light Curve Inversion Techniques”
<i>245th American Astronomical Society Meeting</i>
American Astronomical Society | 2025 |
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“Star Spots with Starry” <i>22nd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun</i> University of California San Diego	2024
“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” <i>Center for Computational Astrophysics Gotham Fest 2023</i> Center for Computational Astrophysics	2023
“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” <i>241st American Astronomical Society Meeting</i> American Astronomical Society	2023
“Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2” <i>237th American Astronomical Society Meeting</i> American Astronomical Society	2021

TEACHING

Adjunct Lecturer	Aug 2025 – May 2026
<i>Baruch College</i>	
– Introductory Physics Lecture	
– Introductory Physics Laboratory	
– Physics II Laboratory	

TECHNICAL SKILLS

Programming	Python, MATLAB, Bash
Scientific Python	NumPy, SciPy, Pandas, Matplotlib, Jupyter Notebook, Astropy, Lightkurve, Astroquery
Computational Methods	Time-Series Analysis, Statistical Modeling, Inverse Problems, Bayesian Inference, Computational Modeling, Spectral Analysis
Software	Git, Linux, L ^A T _E X, Scientific Writing